File formats

This document explains the file formats in src/formats (OFS::Formats) byte by byte, the way the code reads and writes them, along with the odd things found in real files.

We started from Elecbyte's M.U.G.E.N documentation and from how IKEMEN GO (MIT) reads the files. Then every detail below was checked against real files:

- the sample data that comes with Fighter Factory Studio, including its Kung Fu Man (an SFF 2.01 file with 281 sprites and 7 palettes, and an SND with 12 sounds);
- a M.U.G.E.N 1.0 folder with lots of characters, stages, fonts and screenpacks made by other people.

That makes 195 SFF files (75 v1, 114 v2.0, 6 v2.01) with 168,338 sprites, 96 SND files with 17,096 sounds and 70 FNT files. The numbers further down come from these files. Elecbyte's command line tools sff2png.exe (v0.1), sprmake2.exe (2.00beta) and sndmaker.exe (1.02) were used to check our results (see "Checks against Elecbyte's tools" at the end).

None of these files are in the repository. The tests that use them read their folders from OFS_CORPUS_KFM and OFS_CORPUS_MUGEN, and are skipped when those aren't set.

Numbers that take more than one byte are little-endian, and offsets count from the start of the file, unless it says otherwise.

SFF sprite files

sff/SffDocument.h. One class (SffDocument) holds every version. SffDocument::version decides which layout write() uses, and convertTo() changes it.

Version bytes

Bytes 12-15 hold the version as build, minor 2, minor 1, major. So version 2.0.1.0 is stored as 00 01 00 02. These are the ones found in real files:

Stored bytes	Version	Meaning
00 01 00 01	1.0.1.0	SFF v1 (WinMUGEN)
00 00 00 02	2.0.0.0	SFF v2 (M.U.G.E.N 1.0)
00 01 00 02	2.0.1.0	SFF v2.01 (M.U.G.E.N 1.1, adds PNG sprites)

The class keeps the four bytes as they were read (versionBytes) next to the version number.

SFF v1

The header is 512 bytes:

Offset	Size	Field
0	12	"ElecbyteSpr\0"
12	4	version bytes 00 01 00 01
16	4	number of groups

Offset	Size	Field
20	4	number of images
24	4	offset of the first subfile (512)
28	4	size of a subfile header (32)
32	1	palette type: 1 = shared, 0 = each sprite has its own
33	3	unused
36	476	unused, often holds a comment

The images are a chain of subfiles. Each one has a 32-byte header and then the image data:

Offset	Size	Field
0	4	offset of the next subfile
4	4	length of the image data, 0 for a linked sprite
8	2	axis x (signed)
10	2	axis y (signed)
12	2	group
14	2	image number
16	2	for a linked sprite: the index of the sprite it shares
18	1	1 = use the palette of the sprite before ("same palette")
19	13	unused

The image data is an 8-bit PCX file (see PCX below). A sprite with "same palette" set can leave out the PCX palette. It is then drawn with the palette of the last sprite before it that has its own pixels (linked sprites don't count). A linked sprite shows the pixels and the palette of the sprite it points to. Its PCX data is empty, so the next subfile starts right after its header.

In SffDocument, every sprite with its own palette adds one SffPalette (named after the sprite's group and number), and each sprite's paletteIndex points to the palette it's drawn with. That way v1 and v2 files are handled the same.

Odd things found in real files:

- **Wrong lengths.** In 153 subfiles of 8 files (Elecbyte's fight.sff, fightfx.sff, system.sff and Kung Fu Man's intro.sff, plus the intro and ending of one other character) the length is exactly 768 bytes too big. It still counts the palette that was taken out because the sprite uses the palette before it. In the last subfile this even points past the end of the file. So the reader finds the end of the data from the next subfile's offset, and only uses the length when that offset can't be used.
- In the last subfile, the next subfile offset is the file size (never 0).
- The number of groups can't be trusted (one stage stores 0 for 4 groups).
- Two files store 512 as the subfile header size, but the headers are 32 bytes.
- 475 sprites with "same palette" set still have a full palette, which isn't used.
- 1,257 PCX images pad their lines to a multiple of 4 bytes (it's usually 2).
- Lots of files have a comment in the header (60 files) or bytes other than 0 in the unused part of subfile headers (1,755 subfiles).

Writing: subfiles are written in order, and each next offset points right after the data. A sprite whose palette is the same as the one before (same RGB) gets "same palette" and a PCX without a palette. Sprites that came from a file and weren't changed are written with their original PCX bytes, length, link, "same palette" byte and unused bytes. If only the palette of such a sprite changed, the 768 palette bytes at the end of its PCX are replaced (or added) and the image isn't encoded again. The header is written from the 512 bytes that were read, with the counts and offsets updated. The original number of groups and subfile header size stay as they were as long as the groups don't change.

SFF v2 (2.0 and 2.01)

The header is 512 bytes:

Offset	Size	Field
0	12	"ElecbyteSpr\0"
12	4	version bytes
16	8	reserved (0)
24	4	"compatible" version bytes, the same as the version in every file we've seen
28	8	reserved (0)
36	4	offset of the sprite node table
40	4	number of sprites
44	4	offset of the palette node table
48	4	number of palettes
52	4	offset of the literal data block
56	4	length of the literal data block
60	4	offset of the translated data block
64	4	length of the translated data block
68	8	reserved (0)
76	436	unused, some editors put a comment here

Palette node (16 bytes):

Offset	Size	Field
0	2	group
2	2	number
4	2	number of colors (32 or 256 in practice)
6	2	for a linked palette: the index of the palette it shares
8	4	offset of the colors inside the literal block
12	4	length in bytes (colors x 4), 0 for a linked palette

Colors are stored as R, G, B and a fourth byte. Programs don't agree on that byte. Elecbyte's sprmake2 2.00 writes 0. Most palettes we've seen (made with other editors, Fighter Factory's Kung Fu Man too) have 0 for index 0 and 255 for the rest, and a few have other values. SffDocument keeps it in Rgba::a so it can be written back, and ignores it everywhere else (index 0 is see-through, the rest are solid).

Sprite node (28 bytes):

Offset	Size	Field
0	2	group
2	2	number
4	2	width
6	2	height
8	2	axis x (signed)
10	2	axis y (signed)
12	2	for a linked sprite: the index of the sprite it shares
14	1	format, see below
15	1	color depth: 5 or 8 (indexed), 24 or 32 (true color)
16	4	offset of the data inside its block
20	4	length of the data, 0 for a linked sprite
24	2	palette index
26	2	flags: bit 0 set = the data is in the translated block

Format	Name	Data
0	raw	width x height palette indices, nothing in front (raw 24 or 32-bit data is refused: we've never seen it and SprMaker2 can't make it, so its byte order is unknown)
1	(invalid)	never used for data, some programs store it in linked nodes
2	RLE8	4-byte unpacked size, then RLE8 data
3	RLE5	4-byte unpacked size, then RLE5 data
4	LZ5	4-byte unpacked size, then LZ5 data
10	PNG8	4-byte size (width x height), then an 8-bit palette PNG
11	PNG24	4-byte size (width x height x 3), then an RGB PNG
12	PNG32	4-byte size (width x height x 4), then an RGBA PNG

Data in the literal block is used as it is. Data in the translated block is unpacked by the engine when it loads the file (SprMaker2's sprite.decompressonload). Only packed formats show up there in real files (we've seen RLE8 and LZ5). sff2png won't open a file with a raw sprite in the translated block, so the writer never puts new raw data there.

For PNG8 sprites only the PNG's pixel indices are used. The sprite is drawn with the SFF palette its node names, and the PNG's own PLTE is ignored. The writer still puts the SFF palette in the PLTE, so the PNG looks right in other programs.

Every file we've seen (and every file we write) is laid out like this: the palette node table at 512, the sprite node table right after it, then the literal block with all the palette data first (in palette order) and then the sprite data in node order, and last the translated block, which ends at the end of the file.

Links:

- A linked palette or sprite has length 0 and the index of an earlier node. A sprite can link to a sprite that is itself linked.
- The data offset of a linked node isn't used, and files fill it in one of two ways and stick to it. 119 of the 120 v2 files store where the next data would go in the node's block (a "running offset"). Fighter Factory's Kung Fu Man stores the offset of the target's data. SffDocument remembers which one a file uses (linkOffsets) and writes links the same way.
- The format byte of a linked sprite is either 1 or a copy of the target's format. The color depth is usually the target's (9 sprites have a different one).
- A linked sprite has its own palette index, and in 698 sprites it's different from the target's. IKEMEN GO draws the link with its own palette, and so does decodeSprite(). Elecbyte's sff2png does too, except when that palette's size is different from the target palette's (for example a 5-bit sprite shown through a link whose palette has 256 colors). Then it uses the target's palette. Files written by our tools keep link palettes at the size the sprite needs.

Odd things found in real files:

- A font file (num72.sff) says version 2.0.0.0 but has PNG32 sprites. Files like that are treated as v2.01 (version), and the stored version bytes are kept.
- 97 files have text in the unused part of the header ("Made in ... Powered by Fighter Factory").
- 13 RLE8 sprites in one character (SSJ4_GogetaZ2i) end long before all the pixels are there (the data ends in zeros). They're read in lenient mode, with the missing pixels set to 0. Every other sprite reads fine in strict mode.

Writing: palettes and sprites are laid out as above. A sprite that came from a file and wasn't changed keeps its stored bytes, format and depth. A palette keeps its colors, fourth byte included, and links keep their indices and stored fields. With the stored header, this gives back every file we tested byte for byte. New or edited sprites are encoded the way SffCompressionPolicy says: by default RLE8 for 8-bit, LZ5 for sprites that only use indices 0-31 and PNG32 for true color, or else whichever encoding comes out smallest. SFF 2.0 can't store PNG, so those sprites are encoded again, and true color sprites are refused. linkDuplicateSprites and linkDuplicatePalettes turn sprites with the same pixels and palettes with the same colors into links.

Changing the version

convertTo() changes the version:

- v1 to v2: palettes that are the same are merged (v1 stores one for each sprite). The palette of sprite 0,0 (or of the first sprite) becomes palette 1,1, so M.U.G.E.N uses the character's ACT files for it. The PCX sprites are encoded again when saving.
- v2 to v1: a link whose palette is different from its target's becomes a copy, because v1 links always share the palette. True color sprites can't be changed (use adaptToPalette() first).
- 2.01 to 2.0: PNG8 sprites are encoded again, and true color is refused.

Sprite codecs

codecs/. Each decoder takes the packed data (without the SFF size in front) and the number of pixels to make. In DecodeMode::Strict, data that runs out, runs that go past the end of the image

and copies from before its start are errors. DecodeMode::Lenient works like the engines: missing pixels stay 0 and runs that are too long are cut. Extra bytes after the last pixel are ignored.

RLE8

A byte whose two top bits are 01 (0x40-0x7F) starts a run. The low six bits are the run length (0-63) and the next byte is the value. Any other byte is one pixel. So the values 0x40-0x7F always take two bytes, even on their own. The encoder writes runs of up to 63, and single pixels as one byte when it can.

RLE5

For indices 0-31. A packet is:

Bytes	Meaning
1	run length R of the first color
1	bit 7: a first color follows; bits 0-6: number N of extra runs (0-127)
0 or 1	the first color when bit 7 is set, otherwise the first color is 0
N	extra runs, one byte each: bits 5-7 = length - 1, bits 0-4 = color

The first color is repeated R + 1 times (1-256), and each extra run its length (1-8). None of our test files use RLE5, so the decoder was checked against sprmake2 and sff2png.

sprmake2 bug: SprMaker2 2.00beta sometimes writes a packet with 128 extra runs. That number doesn't fit in seven bits and spills into the color flag, so the byte reads as "first color follows, no extra runs" and the rest of the image comes out different from what was meant. Elecbyte's own sff2png reads these packets exactly like we do, so the sprites look the same as in M.U.G.E.N's tools. Our encoder never writes more than 127 extra runs.

LZ5

For indices 0-31. Packets come in groups of up to eight, and each group starts with a control byte. Bit k (lowest first) says whether packet k is a run (0) or a copy (1).

Packet	Bytes	Encoding
short run	1	bits 5-7 = length (1-7), bits 0-4 = color
long run	2	the first byte has bits 5-7 clear and holds the color, the second is length - 8 (8-263)
short copy	1 or 2	bits 0-5 = length - 1 (2-64, never 0), bits 6-7 see below, then distance - 1
long copy	3	first byte: bits 0-5 clear, bits 6-7 = bits 8-9 of distance - 1; then bits 0-7 of distance - 1; then length - 3 (3-258)

A copy repeats the output from distance pixels back (1-256 for short copies, 1-1024 for long ones), and the part it copies from can overlap the part it writes. The top two bits of each short copy are collected, two at a time and highest first, into a spare byte. The first three short copies have their own distance byte. The fourth has none and uses the collected byte as its distance - 1. Collecting starts again after every fourth short copy, and control bytes don't reset it.

The data ends after the last packet. No control byte follows, even when the last packet fills a group of eight (1,096 of the 6,891 LZ5 sprites we've seen end like that, and none has extra bytes).

The encoder finds matches with hash chains over pixel pairs (window 1024, chain depth 48), and picks a run or a copy by how many pixels each byte gives. Once all the data is worked out, it fills in the shared distance bits of each group of four short copies.

PCX

The header is 128 bytes (only the fields we use):

Offset	Size	Field
0	1	manufacturer, 0x0A
1	1	version (5)
2	1	encoding: 1 = RLE, 0 = none
3	1	bits per pixel in each plane (8)
4	8	x min, y min, x max, y max (width = x max - x min + 1)
12	4	horizontal and vertical DPI
65	1	planes: 1 = indexed, 3 = RGB, 4 = RGBA
66	2	bytes per line in each plane (at least the width, padded)
68	2	palette info

RLE: a byte with both top bits set (0xC0-0xFF) is a run of the next byte, byte & 0x3F long. Any other byte is used as it is. Each line holds the planes one after another, each "bytes per line" long, and the padding is dropped after reading. Runs that cross lines are accepted. An 8-bit image can end with 0x0C and 768 bytes of RGB palette. PcxPalette::Optional lets it be missing (SFF v1 sprites with "same palette").

The encoder writes version 5, RLE, one plane, lines padded to an even length, runs that never cross lines, and the palette if asked.

PNG

Through LodePNG (https://github.com/lvandeve/lodepng), in third_party/lodepng. Palette images of any bit depth give back their exact indices: they're read without color conversion and our code unpacks the indices, so palettes with repeated colors keep their indices. Other color types are read as RGBA. Indexed images are written as 8-bit palette PNGs without color conversion, with the PLTE long enough for the highest index. True color images are written as RGB (PNG24) or RGBA (PNG32).

Image files

These are used to bring images and palettes in and out of the editors. PCX and PNG are explained with the sprite codecs above.

GIF

codecs/Gif.h and codecs/Lzw.h. GIF87a and GIF89a files are read with all their frames, and GIF89a files are written (for the GIF export in the Animations editor).

Offset	Size	Field
0	6	"GIF87a" or "GIF89a"
6	2	screen width
8	2	screen height
10	1	bit 7: a global color table follows; bits 4-6: color resolution; bits 0-2: table size n (2^(n+1) entries)
11	1	background color index
12	1	pixel aspect ratio

Then comes the global color table (3 bytes per entry: R, G, B), and then blocks until the end byte 0x3B. The data in blocks is split into sub-blocks: a length byte (1-255) and that many bytes. A length of 0 ends them.

- 0x21 extension: a label byte, then sub-blocks.
    - 0xF9 graphic control, one 4-byte sub-block: flags (bits 2-4 are the disposal method: 0 not given, 1 keep, 2 restore to background, 3 restore to previous; bit 0: the transparent index is used), the delay before the next frame in hundredths of a second (16 bits) and the transparent index. It's for the next image.
    - 0xFF application: an 11-byte name sub-block. NETSCAPE2.0 (or ANIMEXTS1.0) followed by the sub-block 01 lo hi gives the number of loops (0 = forever).
    - 0xFE comment and 0x01 plain text are skipped. A plain text block uses up a graphic control extension that's waiting, as the specification says.
- 0x2C image: left, top, width and height (16 bits each), flags (bit 7: a local color table follows; bit 6: interlaced; bits 0-2: table size), the local table, the smallest LZW code size (one byte) and the LZW data in sub-blocks. Interlaced images store rows 0, 8, 16, ..., then 4, 12, ..., then 2, 6, ..., then 1, 3, 5, ...

LZW. Codes start at the smallest code size + 1 bits and are packed lowest bit first. Code 2^min clears the dictionary, 2^min + 1 ends the data and 2^min + 2 is the first free code. Every code after the first adds an entry: the string before plus the first index of the current one (when the code is the next free entry, it's the string before plus its own first index). Codes get one bit wider when the dictionary reaches 2^width entries, up to 12 bits. When the dictionary is full (4096 entries), encoders either send a clear code or go on without adding entries (a "deferred clear"). We read both. Our encoder starts with a clear code, clears every time the dictionary is full and ends with the end code. The smallest code size has to be 2-8.

Reading: a frame's palette is its local table, or the global one, or a gray ramp if the file has neither. The transparent entry gets alpha 0. Strict mode needs the end byte, and in every frame exactly the pixels it says it has, followed by the end code. Lenient mode keeps the frames read so far and reads a cut off frame as far as its data goes. A 0x0 screen is replaced by the size that fits every frame. Screens and frames bigger than 64 megapixels are refused.

renderGifFrames() puts the frames together the way browsers do. The screen starts see-through, "restore to background" clears the frame's rectangle to see-through (the background color index is

ignored), "restore to previous" puts the screen back the way it was before the frame, and frames are cut to the screen.

Writing: GIF89a with color resolution 7. The global table is written when the document has one. A frame gets its own table when asked, when there's no global table, when its palette is different from the global one (RGB compared) or when it uses indices past the end of the global table. Tables are padded with black to a power of two (at least 2 entries), and the smallest code size is the table's bit count, at least 2. Every frame of an animation gets a graphic control extension. A still image only gets one if it has a delay, a disposal method or a transparent index. The NETSCAPE2.0 block is written when there's a loop count. The screen is made big enough for every frame, because programs handle frames that stick out in different ways (browsers cut them, Pillow makes its canvas bigger).

Checked against Pillow 12. Pillow reads our files to the same pixels (it fills areas that were never painted with the background color, where we leave them see-through). Our decoder reads still, interlaced and animated GIFs written by Pillow (with disposal methods and transparency) to the same pixels Pillow shows.

BMP

codecs/Bmp.h. A 14-byte file header: "BM", the file size (not checked, it's often wrong), 4 reserved bytes and the offset of the pixel data. Then a DIB header, whose first 4 bytes give its size:

Size	Header	Fields
12	BITMAPCOREHEADER (OS/2 1.x)	16-bit width, height, planes and bits per pixel. Color table entries are 3 bytes
40	BITMAPINFOHEADER	signed 32-bit width and height (a negative height means the rows go top to bottom), planes, bits per pixel, compression, image size, horizontal and vertical pixels per meter, colors used, important colors
52 / 56	BITMAPV2/V3INFOHEADER	adds the red, green, blue (and alpha) masks
108 / 124	BITMAPV4/V5HEADER	adds color space fields (ignored)

Compression: 0 = none, 1 = RLE8, 2 = RLE4, 3 = bit fields (a 40-byte header is then followed by three 32-bit masks), 6 = bit fields with alpha (four masks). 4 (JPEG) and 5 (PNG) are refused, and so are the 64-byte OS/2 2.x headers.

The color table comes after the header and masks. Its entries are B, G, R and a reserved byte, as many as "colors used" says, or 2^bits when that is 0 (or too big). Rows are padded to 4 bytes and go bottom to top unless the height is negative. 1, 4 and 8-bit pixels are indices, highest bits first. 16-bit pixels are X1R5G5B5 when there are no bit fields, 24-bit pixels are B, G, R, and 32-bit pixels are B, G, R and a fourth byte when there are no bit fields. Channels from masks of any width are scaled to 8 bits.

RLE8 and RLE4 data is a list of byte pairs. A pair (n, v) with n > 0 is a run of n pixels (in RLE4 the two halves of v take turns). After a 0: 0 ends the line, 1 ends the image, 2 moves the position by the next

two bytes (dx, dy), and n >= 3 is followed by n pixels, padded to 2 bytes. RLE images can't go top to bottom.

Odd things: Microsoft says the fourth byte of 32-bit pixels without compression isn't used, but lots of programs (Pillow too) store alpha there and others store zeros. We use the byte as alpha unless it's 0 in every pixel (Pillow always ignores it). The same goes for an alpha mask. Some programs leave out the padding of the last row, and lenient mode reads those files. Indices past the end of the color table are kept.

Writing: 8-bit images get a BITMAPINFOHEADER and a color table as long as the palette (at least up to the highest index used, at most 256 entries). 24-bit images get a BITMAPINFOHEADER. 32-bit images get a BITMAPV4HEADER with the bit fields 00FF0000, 0000FF00, 000000FF and FF000000 and the sRGB color space, so other programs see the alpha channel. Rows are written bottom to top at 72 dpi (2835 pixels per meter). Pillow reads all three back to the same pixels.

SND sound files

snd/SndDocument.h. A 512-byte header, then a chain of sounds:

Offset	Size	Field
0	12	"ElecbyteSnd\0"
12	4	version bytes (seen: 00 01 00 01, 01 00 01 00, 04 00 00 00)
16	4	number of sounds
20	4	offset of the first sound (512)
24	488	unused (a comment in 87 of 96 files)

Each sound has a 16-byte header and then its data:

Offset	Size	Field
0	4	offset of the next sound (the file size for the last one)
4	4	length of the data
8	4	group (signed)
12	4	sound number (signed)

The data is a whole WAV file, kept exactly as it is. In 206 sounds the RIFF size inside the WAV doesn't match the stored length (both bigger and smaller), which doesn't matter because nothing in the SND relies on it. The sounds are stored one after another in order. Reading and writing gives back every test file byte for byte.

WAV

wav/WavFile.h. RIFF WAVE with chunks (id, 4-byte size, data, and a pad byte after odd sizes). The "fmt " chunk holds the format tag, channels, sample rate, byte rate, block align, bits per sample and maybe an extension (kept as bytes; for WAVE_FORMAT_EXTENSIBLE its sub-format is used). Samples can be unsigned 8-bit, signed 16, 24 or 32-bit PCM, or 32 or 64-bit float. All other chunks

(LIST, "cue ", smpl, fact, ID3 tags and chunks from all sorts of editors, in dozens of different orders in our test files) are kept in order and written back as they were.

By default the reader puts up with the damage that's common in game files: a wrong RIFF size, a last chunk cut short after the samples, junk after the samples, a missing last pad byte, or samples cut short (kept as far as they go). WavParseOptions::strict refuses all of that.

Packed WAV data and other sound files

The audio library (audio/AudioCodec.h) also reads the packed WAV formats that sounds from other tools may use. None of the SND files we tested has them (they're all 8 or 16-bit PCM), so they only matter when importing:

Format tag	Encoding	Layout of the data chunk
0x0006, 0x0007	A-law, mu-law (G.711)	one byte per sample, unpacked to 13 or 14-bit PCM
0x0011	IMA (DVI) ADPCM	blocks of blockAlign bytes: for each channel a 4-byte header (first sample, step index, an unused byte), then groups of 4 bytes (eight 4-bit codes, low half first) for each channel in turn
0x0002	Microsoft ADPCM	blocks: the predictor index of every channel, then their deltas, first samples and second samples (16-bit each). The second sample comes out first, then the 4-bit codes follow, high half first, taking turns between channels
0x0050, 0x0055	MPEG audio (layers I to III)	MPEG frames

ADPCM files give the samples per block in the extension of the format chunk (wSamplesPerBlock, plus the coefficient pairs for Microsoft ADPCM, or else the seven standard ones). A fact chunk gives the exact number of samples and cuts off the padding of the last block.

Sound files are recognized by their first bytes (the extension doesn't matter): RIFF....WAVE; an Ogg page (OggS) whose first packet starts with \x01vorbis (Ogg Vorbis) or \x7FFLAC (Ogg FLAC); fLaC, maybe after an ID3v2 tag; or MPEG audio, maybe after an ID3v2 tag. For MPEG, a good frame header has to be followed, where the frame ends, by another one with the same version, layer and sample rate.

FNT bitmap fonts (version 1)

fnt/FntDocument.h. The header is 32 bytes:

Offset	Size	Field
0	12	"ElecbyteFnt\0"
12	4	version bytes (00 01 00 00)
16	4	offset of the PCX image
20	4	size of the PCX image
24	4	offset of the text

Offset	Size	Field
28	4	size of the text

The image (all the letters side by side) and the text are kept exactly as they are. 17 files have 8 extra bytes between the header and the image (offset 72), and those are kept too. The text has a [Def] section (Size = w,h, Spacing = x,y, Colors = n, Offset = x,y, Type = Fixed or Variable) and a [Map] section with one line for each letter: the letter itself or a 0xNN code, then "x width" for variable width fonts or a letter index (which can be left out) for fixed width fonts. parseFntText() reads both. A ; at the start of a map line is the ; letter and doesn't start a comment.

FNT version 2 fonts are .def text files that point to an SFF, so they aren't part of this format.

Palette files

palette/PaletteFormats.h.

- **ACT** (Adobe Color Table): 256 RGB triplets (768 bytes), maybe followed by a big-endian color count and transparent index (772 bytes). M.U.G.E.N character palettes store the colors backwards: the first triplet is palette index 255 and the last one is index 0. We checked this with sprmake2, whose default (pal.reverseact = 0) puts the last triplet at index 0, and with Kung Fu Man, whose SFF palette 1,1 is kfm.act backwards. ActOrder::Reversed reads and writes that order, and ActOrder::Natural the normal one.
- **RIFF PAL**: RIFF, the size and "PAL ", then a data chunk with a version (0x0300), a color count and four bytes per color (R, G, B, flags). Other chunks are skipped.
- **JASC-PAL** (Paint Shop Pro): the text lines JASC-PAL and 0100, the color count, then R G B for each color. CRLF or LF.
- **ACO** (Photoshop swatches): big-endian. A version 1 part (version, count, then for each color a color space and four 16-bit values), maybe followed by a version 2 part that repeats the colors with UTF-16 names (a 32-bit length in code units, counting the 0 at the end). RGB, HSB, CMYK, Lab and grayscale are converted. We write both parts, with RGB values times 257 (so reading them back with >> 8 gives the same value) and the palette index as the name. A file can end after its version 1 part.
- **ASE** (Adobe Swatch Exchange): big-endian. "ASEF", version 1.0 (two 16-bit numbers) and a 32-bit block count, then blocks of a 16-bit type and a 32-bit length: 0x0001 color, 0xC001 group start (with a name), 0xC002 group end. A color holds a 16-bit name length in UTF-16 code units counting the 0 at the end, the UTF-16 name, a 4-letter color model ("RGB ", CMYK, "LAB ", Gray), its 32-bit float values (RGB, CMYK and gray from 0 to 1; Lab: L from 0 to 1 for 0-100, then a and b) and a 16-bit color type (global, spot, normal). Groups are flattened and unknown models are skipped. We write plain RGB colors named after their index.
- Palettes can also be taken from PCX, palette PNG and 1, 4 or 8-bit BMP images (the color table in order) and from GIF images (the global color table, or the first frame's local table, with all its entries).
- readPaletteFile() recognizes PNG, GIF, BMP, RIFF PAL, JASC-PAL and ASE by their first bytes, then ACT by its size (768 or 772 bytes), PCX by its first byte and last of all ACO. A 768 or 772-

byte file that starts like an image but can't be read as one is read as ACT, because the first colors of an ACT file can spell anything.

SprMaker2 and SndMaker input files

maker/SprMakerDef.h and maker/SndMakerScript.h. Text files for Elecbyte's command line tools. The SprMaker2 export in the Sprites editor writes a .def next to PNG images and ACT palettes, and a SndMaker list does the same for the WAV files of an SND. The syntax below comes from the sample files that come with M.U.G.E.N 1.0, and we checked it by running sprmake2.exe (2.00beta) and sndmaker.exe (1.02) on changed copies of them.

SprMaker2 .def. A ; starts a comment anywhere on a line. Section names and keys can be upper or lower case.

```
[Output]
filename = chars/kfm/kfm.sff       ; required
[Option]
input.dir = ...                    ; folder of the images (default: the .def's folder)
sprite.compress.5 = lz5            ; 32-color sprites: none, lz5, rle5
sprite.compress.8 = rle8           ; 256-color sprites: none, rle8
sprite.compress.24 = none          ; true color: none
sprite.decompressonload = 0        ; 1: data in the "translated" block
sprite.detectduplicates = 0        ; 1: also link sprites with the same bytes (0: same names only)
sprite.autocrop = 0
pal.detectduplicates = 0
pal.discardduplicates = 0
pal.reverseact = 0
pal.reversepng = 0
[Pal]
1,1, kfm.act, 0,255                ; group, number, file, first color, last color
[Option]
sprite.usepal = 1,1                ; -1: add and use the image's own palette
sprite.removecolors = -1           ; a range like 32,255 lowers the color depth
[Sprite]
0, 0, stand00.png, 18, 105         ; group, number, file, axis x, axis y
0, 1, stand01.png, 18, 104 ? usepal = -1 ? crop = 0,0,39,47
```

[Option] sections can be anywhere, and they change the settings for the sprites listed after them. After a ? a sprite line can set usepal and crop (the corners are included, and the axis moves with the crop) just for itself. More of them are separated by more ? marks. Any other key, section or value is refused ("Invalid option"). sprmake2 still exits with code 0, so whoever runs it has to check that the output file is there. What we saw:

- With pal.reverseact = 0, ACT files are read in M.U.G.E.N order (the last triplet is index 0). The first and last color of a [Pal] line pick palette indices after the order is turned around, so a 32-color palette is g,n, file.act, 0,31 with the colors in the last 32 triplets.
- The palettes of the new file are the [Pal] entries in order, then the palettes added for sprites with usepal = -1 (named after the sprite).
- A sprite's color depth has to match its palette. 256 colors give 8-bit sprites and 32 colors give 5-bit ones (made from 8-bit images with sprite.removecolors = 32,255). A 16-color palette gives a raw 4-bit sprite, and other sizes are refused.

- Lines with the same file name are linked. Sprites are linked even when they use different palettes, and palettes even under different numbers.

SprMakerDef holds the palettes and the sprites, and each sprite has the settings that apply to it. So a file read with options on its lines and [Option] sections all over writes back as the same thing (with [Option] sections where the settings change). exportSprMaker() turns an SFF into such a project. Every palette becomes a [Pal] entry with an ACT file in M.U.G.E.N order (padded to 32 or 256 colors). Every indexed sprite becomes an 8-bit PNG group_number.png, and its palette comes from sprite.usepal (with sprite.removecolors = 32,255 for 32-color palettes). Linked sprites and palettes name their target's file, the decompress on load flag is kept, and autocrop and duplicate checks are off. True color sprites (sprmake2 can't store them), sprites without a palette and repeated sprite numbers are left out with a warning. Repeated palette numbers get the next free number.

SndMaker list. SndMaker reads its input from the console (sndmaker < sound.txt). The first line names the SND file to make, then each sound takes three lines: the WAV file, the group and the sound number. An empty line or the end of the input ends the list. File names are used exactly as written (spaces at the start and end too) and from the folder SndMaker runs in. Numbers can have spaces around them and are 32-bit signed (a missing last number reads as 0, but our reader refuses it). SndMaker writes the version bytes 04 00 00 00, an empty comment and every WAV file as it is, which is exactly what SndDocument::write() makes with those version bytes. exportSndMaker() writes each sound's data as group_number.wav (with _2, _3 and so on for repeated numbers), and can put a folder in front of the names in the list.

The Fighter Factory export format (.ffe) isn't supported. Its layout isn't documented and we had no sample.

Resize filters

image/Resize.h. The Resize dialog has Nearest Neighbor, Bilinear, EPX + Nearest neighbor, EPX + Bilinear, HQ + Nearest neighbor and HQ + Bilinear. The EPX and HQ filters first make the image 2, 3 or 4 times bigger, whichever is the smallest that reaches the bigger of the two scale factors (4 for anything above that, nothing when the image doesn't get bigger). Then nearest neighbor or bilinear gets it to the exact size. Nearest neighbor and EPX keep indexed images indexed. Bilinear and HQ work in RGBA, and the result goes back to the image's palette with adaptToPalette() unless RGBA was asked for.

- EPX is Scale2x at 2x, Scale3x (AdvMAME3x) at 3x and Scale2x twice at 4x. At the edges the border pixels repeat. Fully see-through RGBA pixels count as the same whatever color they hold.
- HQ uses the hqx library (https://github.com/grom358/hqx, LGPL-2.1+). Its hqxInit() would fill a 64 MB table from RGB to YUV. We only fill the entries for the colors in the image before each call, so the rest of the table never takes up memory. Solid images go through hqx once and come out exactly as the library makes them. With transparency it takes more work. The see-through pixels get one color that hqx sees as different from every visible color (so edges against the background are found), and the channels are blended with premultiplied alpha so that color never shows. This works because hqx decides only from RGB and its weights are multiples of 1/16, so the alpha byte can carry any 4-bit value times 16 through the filter and come out as its

exact weighted average. Two runs per value (high and low half) for alpha and for each premultiplied channel give exact sums, eight runs in all.

Checks against Elecbyte's tools

The oracle tests (Windows only, with OFS_CORPUS_MUGEN set to a M.U.G.E.N 1.0 folder) run:

- sff2png <file> <prefix>: saves every sprite of an SFF v1 or v2.0 file as a PNG, plus a SprMaker2 .def with the group, number, file and axis of each (the output folder has to exist). It can't open SFF 2.01 files. We write v1 and v2.0 files with every format that fits (raw, RLE8, RLE5, LZ5, PCX), linked sprites and palettes and odd sizes, and compare sff2png's pixels and palettes with ours. They all match.
- sprmake2 <def>: makes an SFF from the PNG or PCX images listed in a .def. We save our decoded sprites as PNGs, write the .def (with sprite.usepal, and sprite.removecolors = 32,255 for 5-bit sprites, which SprMaker2 only makes from a 32-color [Pal] entry), and read the SFF it makes. RLE8 and LZ5 sprites (all 279 indexed Kung Fu Man sprites too) come back pixel for pixel. So do RLE5 sprites, except the ones hit by the RLE5 bug above, and those read exactly the way sff2png reads them. Our writer also gives back the SFF files sprmake2 makes byte for byte.
- sprmake2 <def> on our SprMaker2 export (exportSprMaker()) of an SFF v2.0 test file (48 sprites, 6 linked, 4 palettes, one linked), the indexed sprites of an SFF v2.01 test file, an SFF v1 test file, and Kung Fu Man (279 indexed sprites, 41 linked, 7 palettes). sprmake2 takes every .def we make, and the SFF it makes has the same palettes (numbers, order, colors, links) and sprites (numbers, order, axes, pixels, palettes, decompress on load flags, links). Our writer gives back those files byte for byte too.
- sndmaker < list on our SndMaker export of a test SND and of Kung Fu Man's sounds. Every sound comes back with the same group, number and WAV bytes, and the file is the same, byte for byte, as what SndDocument::write() makes with SndMaker's version bytes.

What we learned from the tools: sff2png ignores a link's palette when its size doesn't fit (see Links above), raw sprites in the translated block make it refuse the file, and SprMaker2 2.00 writes 0 as the fourth palette byte and turns ACT files around by default.